

Homework 1: Music Release Decade and Market

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This experiment classifies release decade and release market from the supplied 30-second audio excerpts. It first trains classifiers on frozen MERT features, then fine-tunes the last four MERT blocks. Qwen2.5-Omni provides an audio language model comparison. All three methods are evaluated on the same official validation recordings.

Dataset	Classification task	Train	Validation	Test
dataset_A	US music, six decades: 1960s to 2010s	1026	132	132
dataset_B	1980s music, six release markets	798	102	102

Audio is mono, 24 kHz, PCM 16-bit WAV. Each class has 171 training and 22 validation recordings in A, and 133 training and 17 validation recordings in B. The assignment states that the official splits are artist-disjoint. No additional labeled audio was used.

Test labels are hidden. Accuracy refers to official validation, except for the internal training holdout on page 4. Top-3 is correct when the true label is among three distinct ranked predictions.

Code: <https://github.com/Jimmynycu/music-era-market-classification>

Public course submission folder: <https://drive.google.com/drive/folders/1mpd0TqWgGYK-VmeExzmrSfgONGZU-20i>

Frozen MERT features and classifiers

train.py learns six-class linear weights from fixed features extracted by features.py; MERT stays unchanged.

features.py splits each 30-second recording into six non-overlapping 5-second windows. Each window is centered and divided by $\sqrt{\text{variance} + 1e-7}$. Frozen MERT-v1-95M returns 13 hidden-state levels with 768 channels each. The mean and population standard deviation over all frames in all six windows give 1,536 features per level.

The program also extracts 244 acoustic features from the full excerpt: time means and standard deviations of 64 log-mel bands, 20 MFCCs, 20 MFCC deltas, 12 chroma bins, and centroid, bandwidth, rolloff, flatness, RMS and zero-crossing rate. STFT uses 2,048 samples and a hop of 512. Features are cached for train.py.

Model selection uses stratified three-fold CV within the training split (seed 2026). StandardScaler is fitted on the training part of each fold, followed by per-recording L2 normalization. Five fixed feature recipes each use Ridge with $\alpha = 0.01, 0.1, 1, 10$ or RBF SVC with $C = 0.1, 1, 10, 100$ ($\gamma = \text{scale}$): 40 candidates per task.

Task	Selected features	Size	Classifier
Release decade	Mean levels 1-4, 5-8 and 9-12, concatenated with acoustic features	4,852	Ridge $\alpha=1$
Release market	Mean of levels 5-8	1,536	Ridge $\alpha=1$

Selection maximizes Top-1 + 0.5×Top-3; ties use Top-1, then fixed candidate order. The scaler and classifier are refitted on all official training data before validation. No augmentation is used.

Frozen classifier validation results

All confusion matrices show counts. Rows are true classes; columns are predicted classes.

Release decade: Top-1 40.2%, Top-3 83.3% (n=132)

True class	1960s	17	3	0	1	0	1
	1970s	7	8	2	2	2	1
	1980s	1	2	11	3	5	0
	1990s	1	1	7	5	6	2
	2000s	4	2	2	2	4	8
	2010s	2	2	1	2	7	8
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted class					

47/79 errors (59.5%) are between adjacent decades. The 2000s and 2010s are confused 15 times. Recall is 17/22 for the 1960s and 4/22 for the 2000s. Recording style may carry decade cues, but this experiment does not establish which cues the classifier uses.

Release market: Top-1 41.2%, Top-3 75.5% (n=102)

True class	US	5	5	1	1	2	3
	UK	5	7	0	2	1	2
	Brazil	0	0	12	1	1	3
	Spain	0	5	5	4	1	2
	Germany	2	2	2	1	7	3
	Italy	4	4	1	0	1	7
		US	UK	Brazil	Spain	Germany	Italy
		Predicted class					

US and UK account for 10/60 errors (16.7%). Recall is 12/17 for Brazil and 4/17 for Spain. These labels describe release markets, not artist nationality or vocal language. Different markets may share musical styles and production practices.

Acoustic baseline Top-1 / Top-3: A 31.8% / 72.0%; B 22.5% / 63.7%. MERT-only A: 41.7% / 79.5%. The CV-selected A model has higher Top-3, but not the highest Top-1.

Partial fine-tuning of MERT

`finetune_model.py` defines the model; `finetune.py` handles data splits, training and checkpoints.

The same pretrained MERT base is loaded. Transformer blocks 8–11 (zero-based) are unfrozen: 28,351,488 pretrained parameters. Other weights stay fixed. Final-layer frame means and $\sqrt{\text{population variance} + 1e-5}$ form a 1,536-dimensional vector, followed by LayerNorm(1536), Dropout(0.1) and Linear(1536,6).

Each training recording contributes one random 5-second crop per epoch, with the same waveform normalization as the frozen method. Training uses FP32 cross-entropy and AdamW: backbone learning rate $1e-5$, head rate $1e-3$, weight decay 0.01 and gradient norm cap 1. Microbatch size is 4, accumulated to 16. Validation and test average logits from six fixed 5-second windows.

A stratified 20% holdout is drawn only from official training data (seed 2026). Epochs 1–6 are compared by Top-1 + 0.5×Top-3, breaking ties by Top-1 then earlier epoch. Holdout evaluation also covers all 30 seconds.

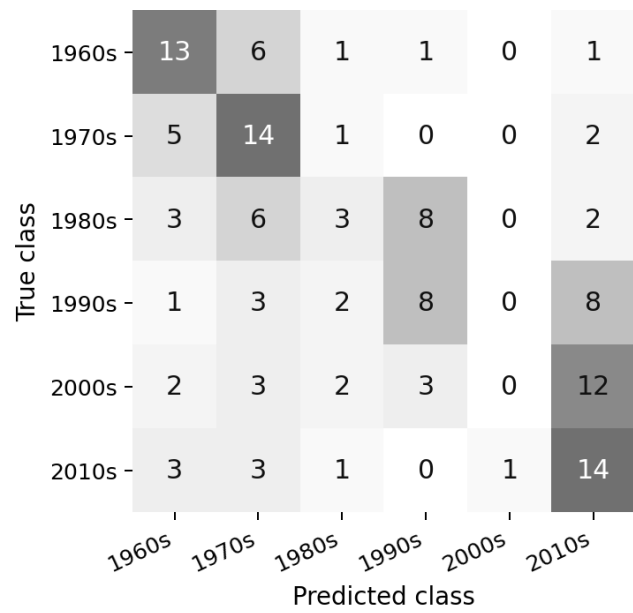
Task	Internal fit / holdout	Epochs	Holdout Top-1	Holdout Top-3
Release decade	820 / 206	6	39.8%	78.2%
Release market	638 / 160	3	26.9%	60.6%

After epoch selection, the original pretrained weights and a fresh head are reloaded, then fitted on all official training data. Records confirm nonzero gradients and weight updates in all four blocks, with frozen weights unchanged. Each task stores its updated blocks and head.

Fine-tuned model validation results

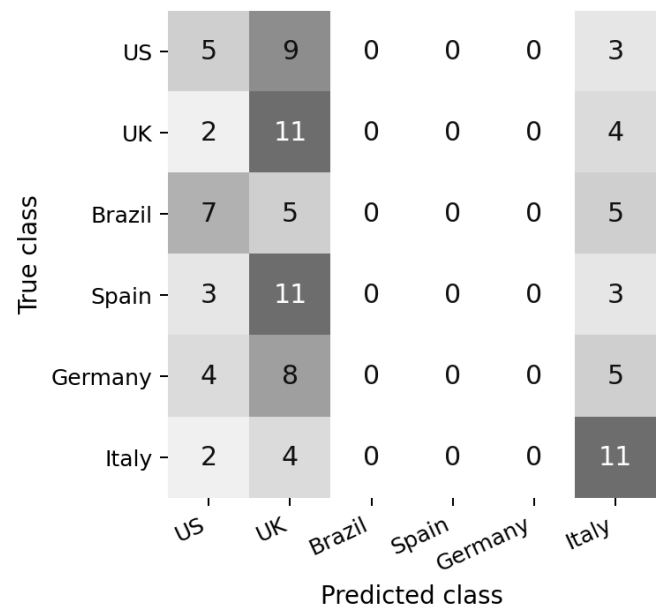
Top-1 and Top-3 are lower than the frozen classifiers for both tasks.

Release decade: Top-1 39.4%, Top-3 79.5% (n=132)



Adjacent-decade errors: 44/80 (55.0%). Recall for the 2000s is 0/22; only one of 132 recordings is predicted as 2000s.

Release market: Top-1 26.5%, Top-3 53.9% (n=102)



Brazil, Spain and Germany are never predicted as Top-1, so recall is zero for all three. Predictions are confined to US, UK and Italy.

This fine-tuning experiment was defined after observing official validation results of the frozen model. Fitting and epoch selection use only official training data. The methods also differ in pooling and classifier head, and frozen A includes acoustic features, so this comparison cannot isolate the effect of unfreezing.

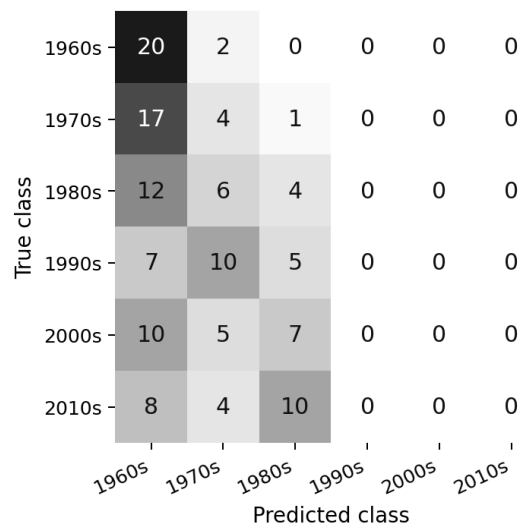
Audio language model comparison

Qwen2.5-Omni-3B Thinker, Q4_K_M language model and Q8_0 audio projector; llama.cpp b11049 / Vulkan.

alm_eval.py sends the full 30-second clip to a local server. It is resampled to 16 kHz, padded with 30 seconds of silence, and processed as two 3,000-frame chunks (1,500 audio tokens). This experimental path has not been shown equivalent to original Transformers preprocessing.
The fixed prompt asks for six ranked labels using instruments, rhythm, voice and production cues from the whole clip. B explicitly separates release market from nationality and language.

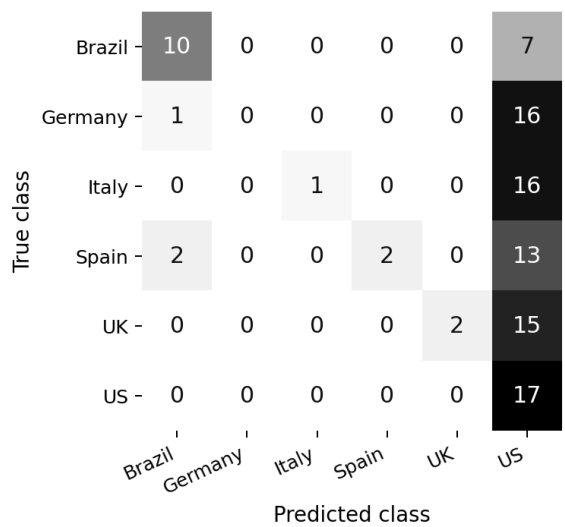
One preset prompt was chosen to limit inference cost and parsing ambiguity; it was not changed after validation. Exact prompts and label mappings: alm_eval.py. GBNF requires each of six codes once; parsing takes the top three. Scores 6 to 1 are ranks. Temperature=0, seed=42. An invalid answer is retried once with the same input, then replaced by a flagged fixed ranking. Server errors stop the run.

Release decade: Top-1 21.2%, Top-3 62.9% (n=132)



Never Top-1: 1990s, 2000s, 2010s. Most common: 1960s (74/132).

Release market: Top-1 31.4%, Top-3 61.8% (n=102)



Never Top-1: Germany. Most common: US (84/102).

Every validation recording was evaluated: A 132, B 102. All 234 outputs were valid; no fallback rankings were used.

Running the code and sources

features.py extracts features; train.py fits the frozen-feature classifiers; finetune.py trains the fine-tuned models. Follow README to install packages and set the A/B test paths, then run inference.py. README contains the full commands and model paths.

Write D15921B14.json with three distinct classes per recording, ordered by descending model score. The default is the fine-tuned model; README also gives the frozen-model command. verify_submission.py checks the output format and reproduced predictions.

Both methods were rerun from all 234 test audio files and exactly reproduced the saved predictions. Test labels are hidden, so this verifies reproducibility, not test accuracy.

Pretraining overlap with these recordings is unknown. Anonymized data has no artist names, so internal training splits cannot be grouped by artist. The causes of class concentration in the audio language model were not separately tested.

Sources (click the names)

Models and papers: [MERT paper](#); [MERT weights / code](#); [Qwen paper](#); [Qwen model](#); [GGUF weights](#); [llama.cpp](#); [audio runtime caveat](#)

Software: [PyTorch](#); [Transformers](#); [Hugging Face Hub](#); [librosa](#); [scikit-learn](#); [NumPy](#); [SciPy](#); [SoundFile](#); [joblib](#); [threadpoolctl](#)

Figures, report and data: [Matplotlib](#); [ReportLab](#); [Course dataset / instructions](#)